SERMA
SAFETY & SECURITY

ALPhANOV
Optics & Lasers Technology Center

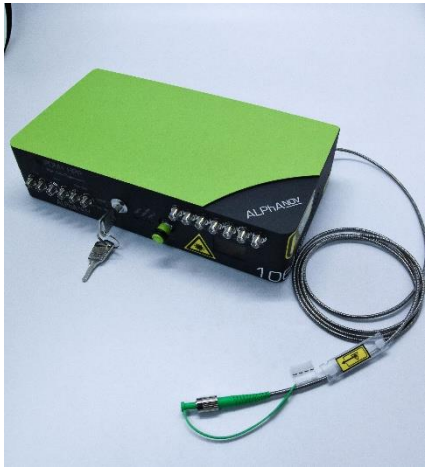


Analyse de l'efficacité des lasers pico et femto seconde à l'injection de faute

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LASER TECHNOLOGY



Nanosecond laser source



Picosecond laser source



Femtosecond laser source

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NANOSECOND LASER SOURCE

- Single-shot, burst mode or continuous
- Up to 250 MHz repetition rate
- Up to 1.8 W peak power
- Delay trigger to pulse : 70 ns typical
- Pulse width down to ~1 ns
- Wavelength : 1064 nm
- Jitter : 20 ps typical

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PICOSECOND LASER SOURCE

- Single-shot to 10 MHz
- Delay trigger to pulse : 120 ns typical
- Pulse width : 30 ps typical
- Wavelength : 1064 nm
- Jitter : <100 ps

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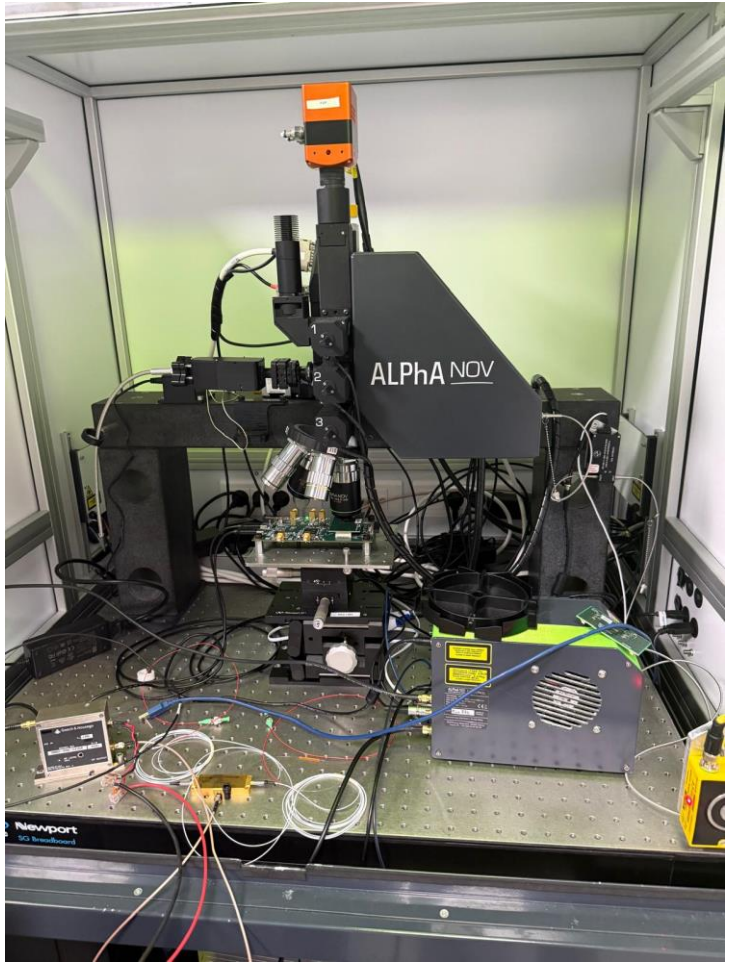
FEMTOSECOND LASER SOURCE

- Single-shot to 40 MHz
- Delay trigger to pulse : 500 ns
- Pulse width : 600 fs
- Wavelength : 1560 nm
- Jitter : 25 ns

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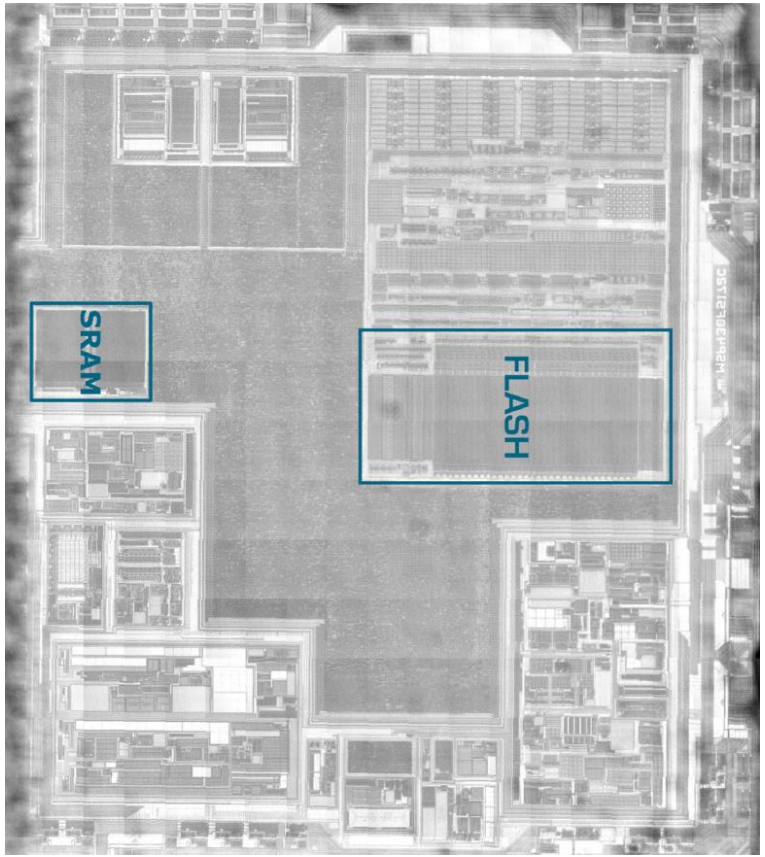
METHODOLOGY MSP430

- Embedded code development
  - Flash memory single address read
  - Synchronization signal when read operation is performed
- Single address location identification
  - EMMI setup
  - Looped write operations on target address
- Embedded Flash memory scan
  - Position
  - Timing
- Read operations transient fault model
  - Trigger signal from the DUT
  - Synchronized laser shot
  - Programmed Flash cells (read = 0)
  - Targeted bit is read as 1
- Analysis of results



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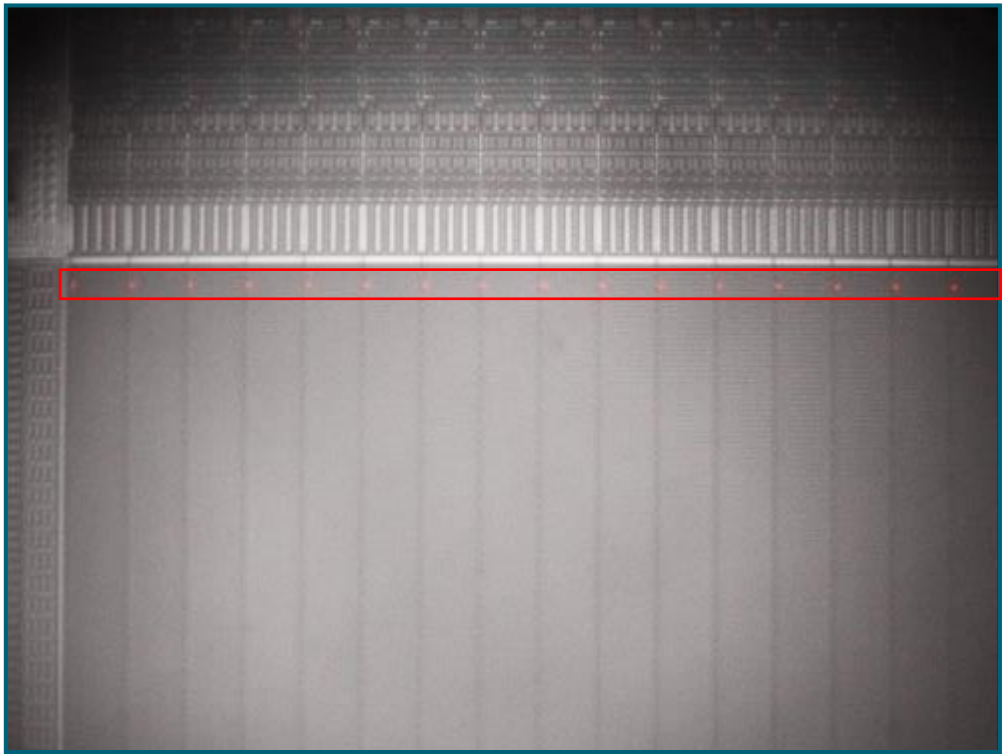
PHOTON EMISSION ANALYSIS ON MSP430



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PHOTON EMISSION ANALYSIS ON MSP430

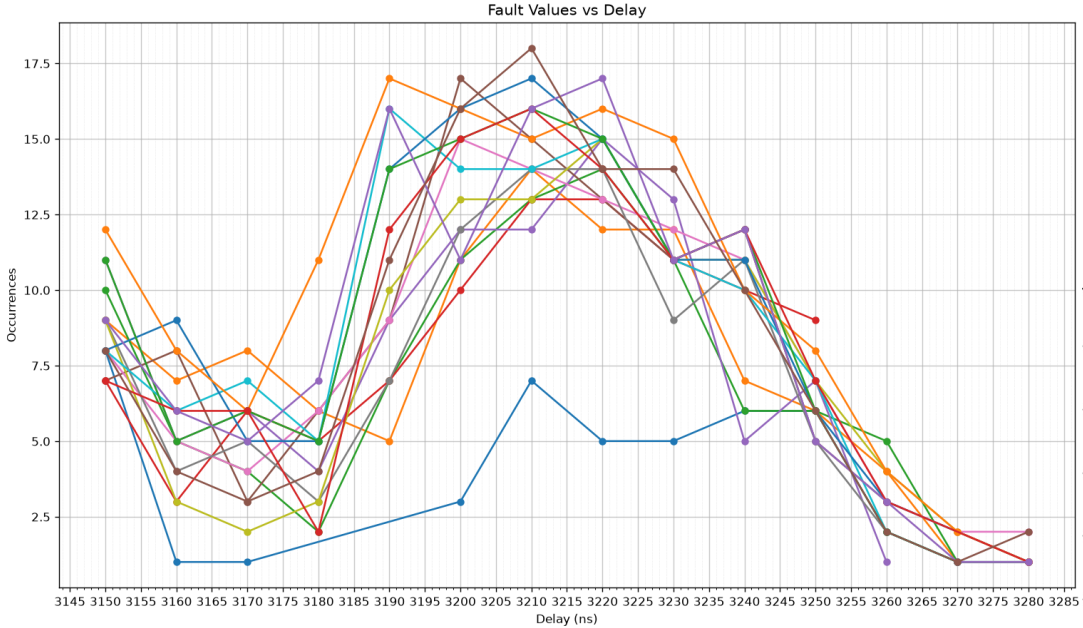
- Each emission site is a bit in the flash memory
- 16 points are visible, indicating a 16-bit memory

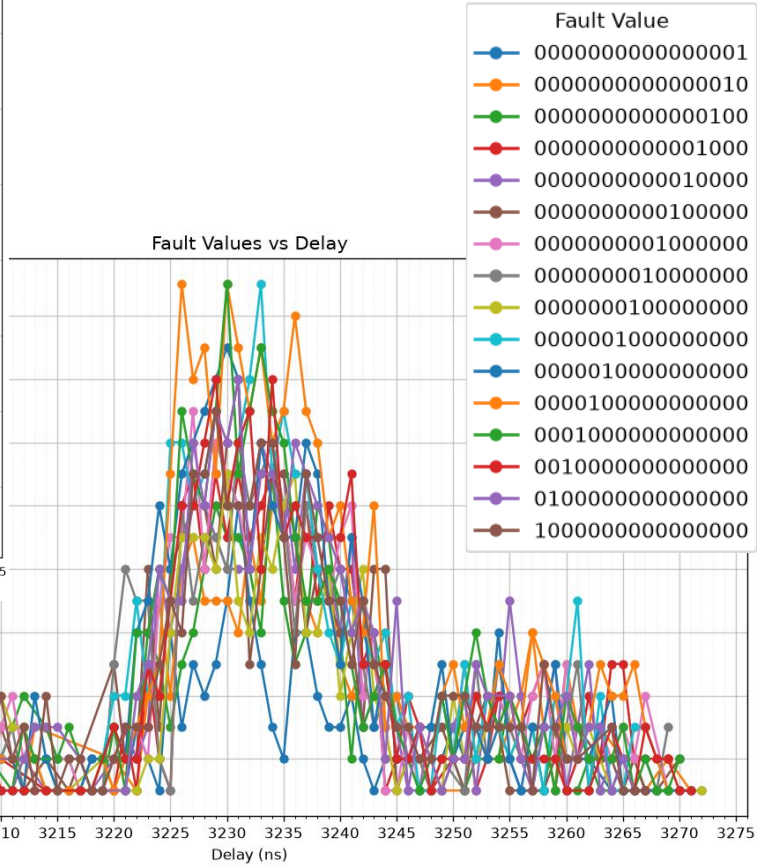


Photon emission analysis with 20x objective

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RESULTS SOURCE NANO VS PICO - TIMING

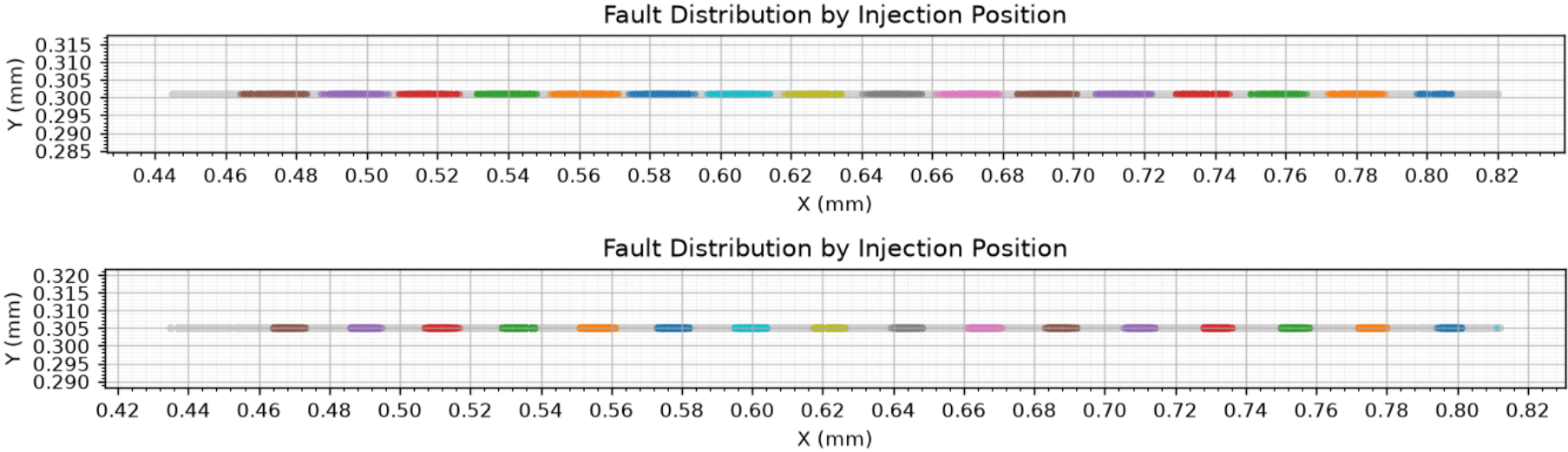




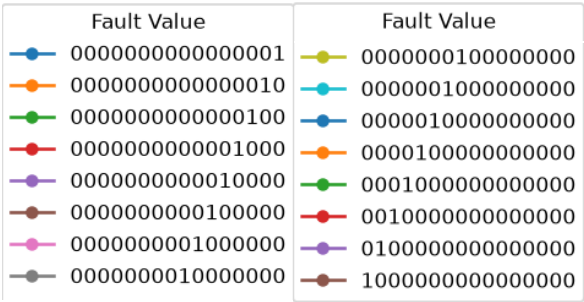
- Tighter sensitivity interval using Picosecond laser source
  - ~20 ns vs over 100 ns

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RESULTS SOURCE NANO VS PICO - POSITION



- Smaller effect area using Picosecond laser source
  - ~12 µm vs ~40 µm



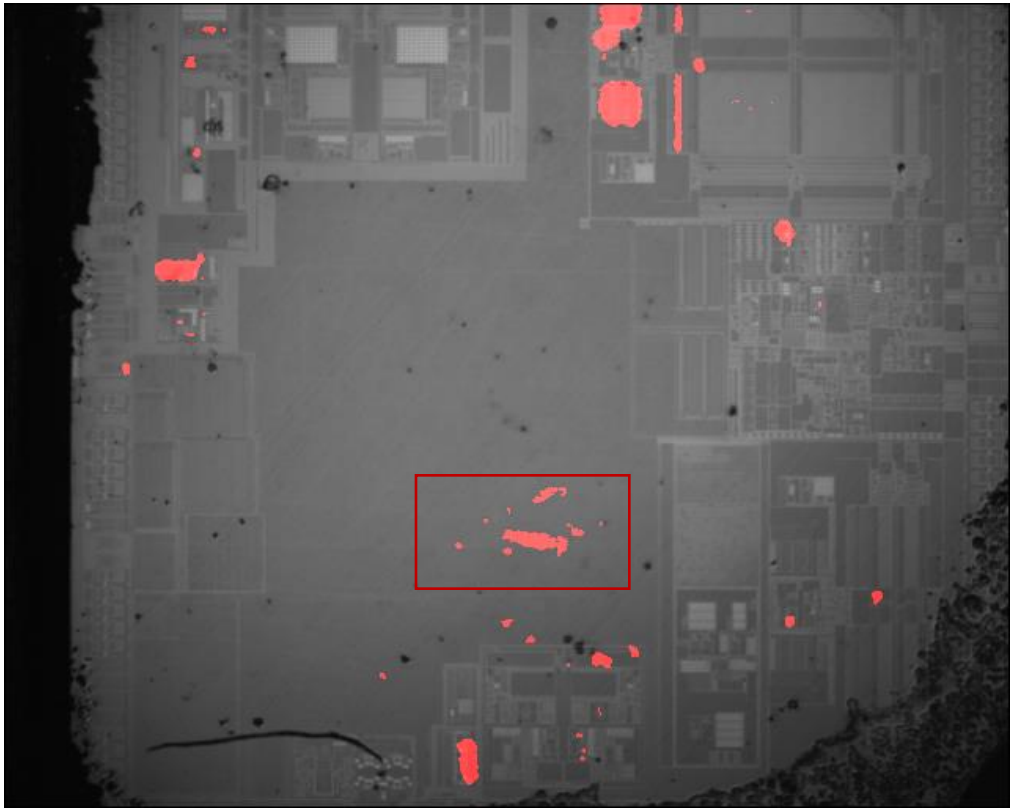
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METHODOLOGY MICRO-CONTROLLER

- Software development
  - Conditional Branch (10 successive conditional test)
  - Software instrumentation
- Identification of the physical location in logic part
  - EMMI setup
- Scanning timing and area identified
  - Finding configuration with laser nano
- Analysis of results

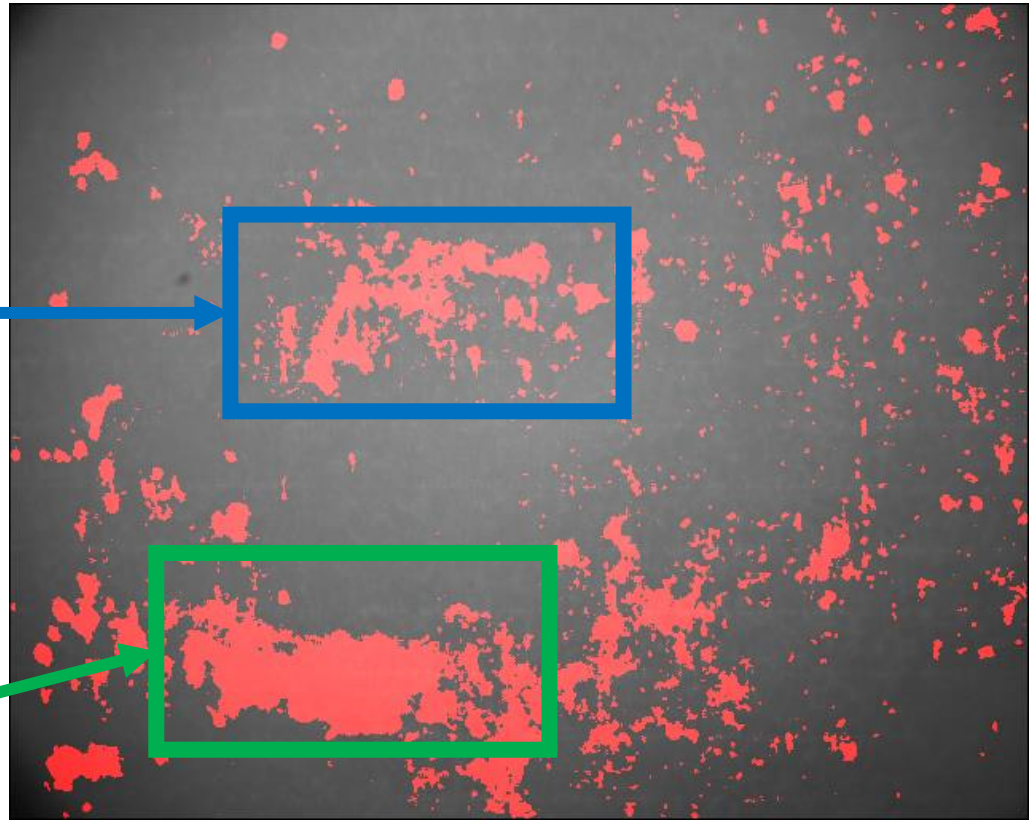
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PHOTO EMISSION WITH OBJECTIVE 2.5X



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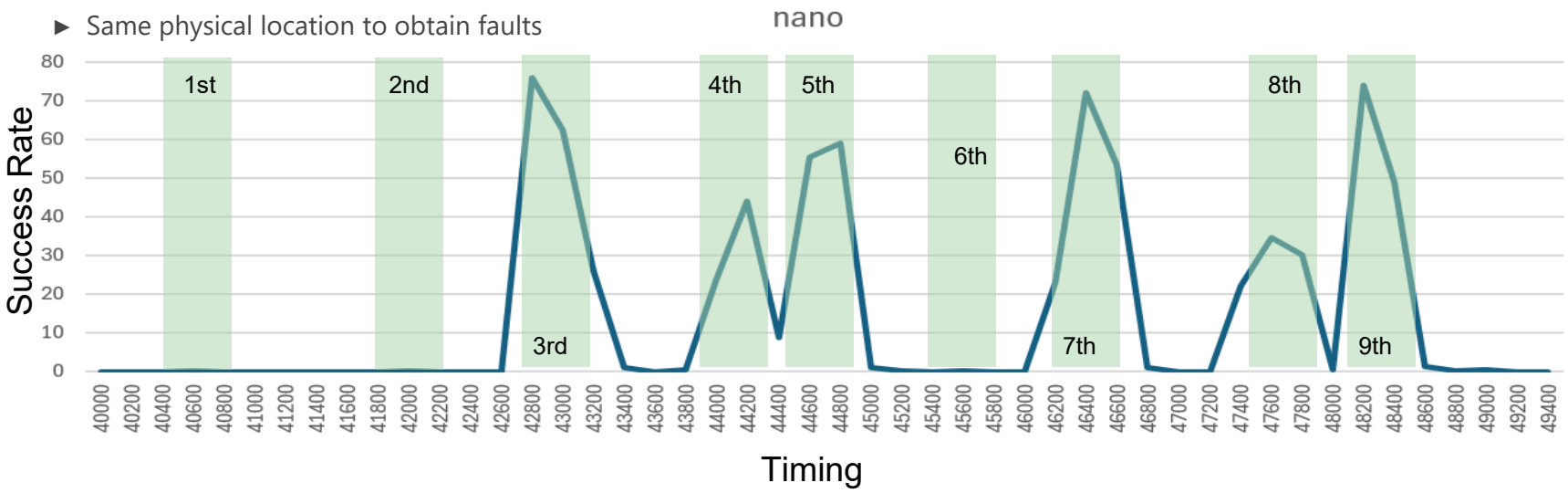
PHOTO EMISSION WITH OBJECTIVE 20X



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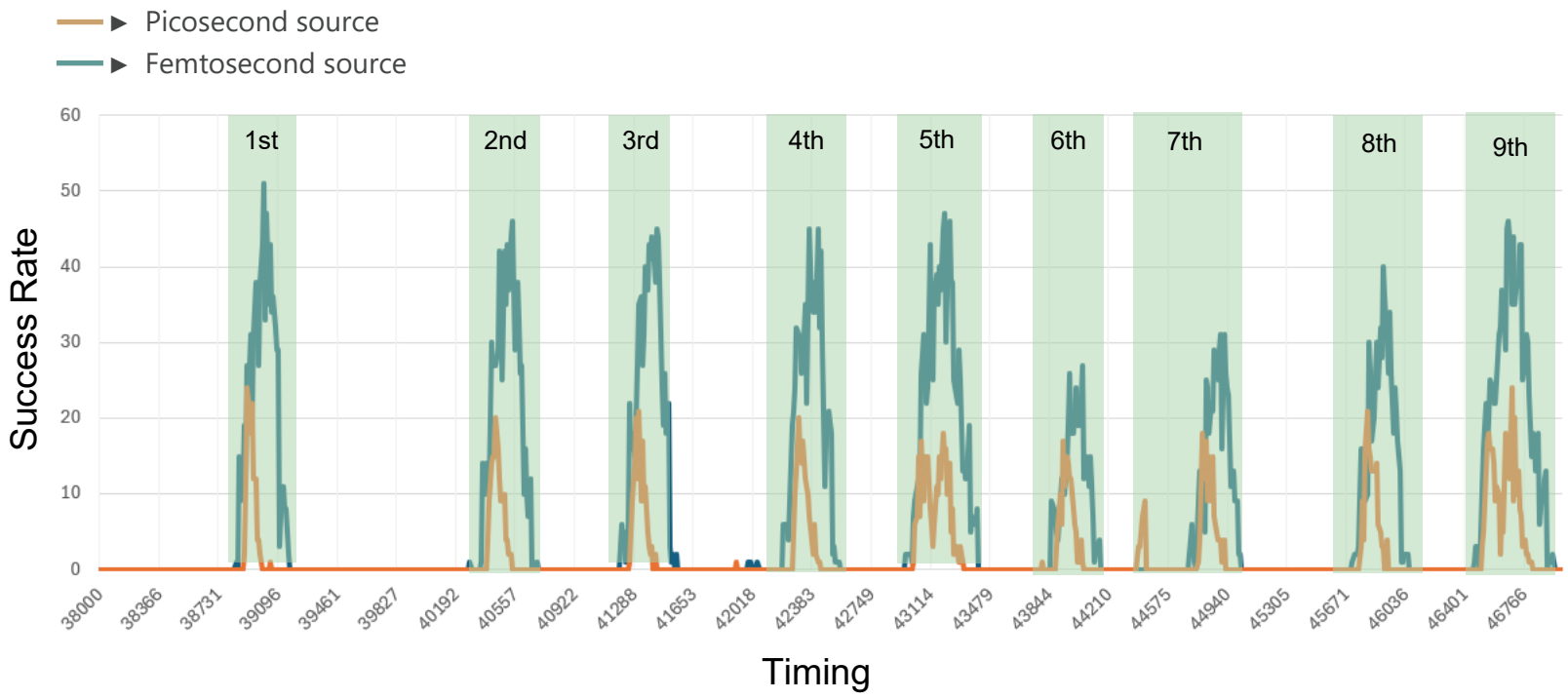
RESULTS – NANOSECOND LASER SOURCE

- Faulted values obtained for laser setup
- Same physical location to obtain faults



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RESULTS – PICO VS FEMTO LASER SOURCES



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METHODOLOGY ON A SECURE ELEMENT

- Same C source code loaded
- Micro-controller with hardware security mechanism such as laser sensors
- Efficiency of the laser sensor?

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RESULT

Laser Nano-second

- Fault produced but Injection detected every time
- With laser with low energy (no laser detection)
  - No fault can be produced

Laser Pico-second

- Fault produced not detected everytime
- Fault not detected for 1.70%
- Usage of an attenuator
  - Best success rate of expected faults
  - Fault not detected for 3.20%

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CONCLUSION

- Pico and Femto laser sources allow to produce faults!
- To reduce time for test campaign, EMMI is mandatory
- Femto seems to have better efficiency for faults produced in the logic part
- To reduce peak instability, an attenuator is necessary
- Pico can produce faults without being detected by sensors.
- Femto? Test campaign yet to be done!

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QUESTIONS?



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